Six math clues · answer key

How Animals Get Ready for Winter

Add the same number once for each group or row. Welcome a drawing of dots as an explanation.

Math 1 · 12

4 + 4 + 4 = 12, so there are 12 seeds.

Math 2 · 10

5 + 5 = 10, so there are 10 pinecones.

Math 3 · 16

4 + 4 + 4 + 4 = 16, so there are 16 leaves.

Math 4 · 25

5 + 5 + 5 + 5 + 5 = 25, so there are 25 bees.

Math 5 · 8

2 + 2 + 2 + 2 = 8, so there are 8 air holes.

Math 6 · 15

5 + 5 + 5 = 15, so there are 15 tracks.

Six equal-group and array problems use rows and columns from 2 to 5 and repeated addition. The numbers are for practice, not real animal counts.

Reading & writing · answer key

How Animals Get Ready for Winter

Reading · 1

Ice forms in the frog's body but outside its cells, and glucose (sugar) from its liver helps protect the cells. When it thaws, its heart starts first.

Reading · 2

A chipmunk saves energy by letting its body get cold in torpor, with short warm-ups; many mammal hibernators use brown fat to warm back up. Honeybees do not hibernate: they huddle in a cluster and the inside bees shiver their flight muscles.

Reading · 3

About six inches of snow works like a blanket, so the space under it stays near 32 degrees Fahrenheit. Scientists think day length starts a hare's coat change, so if snow comes late, a white hare can sit on brown leaves.

Writing

Accept a spoken, dictated, drawn, or written sentence that tells how one animal gets through winter.

Wood frogs: ice forms outside cells while liver glucose protects them; on thawing the heart returns first. Chickadees cache food and use the hippocampus to find it; one study found more new neurons there in fall, but later studies disagree and the cells seem to replace old ones, so we never say they grow bigger brains. In most small mammal hibernators, hibernation is long torpor bouts with brief rewarming, often fuelled by brown fat; a bear cools only a few degrees. With no brood, honeybee clusters keep the core above about 64 F. Day length is thought to trigger the hare molt (about 40 days). About 6 in of snow keeps the subnivean zone near 32 F. Woolly bears make glycerol; we skip the stripe folklore. Watch wildlife from a distance; use warm, not hot, water.

Offer dictation: let the child say a complete thought while you write it, then invite them to add a picture, label, or sentence.

Four winter clues

How Animals Get Ready for Winter

Quiz 1. What protects a frog's cells?

Glucose, a sugar from the frog's liver, helps protect its cells while ice forms outside them.

Quiz 2. Why does a chipmunk cool down?

In torpor its body gets colder and slower, which saves energy.

Quiz 3. How do inside bees make heat?

The inside bees shiver their flight muscles without moving their wings.

Quiz 4. Why is it warmer under snow?

About six inches of snow insulates the space, so it stays near 32 degrees Fahrenheit.

Wood frogs: ice forms outside cells while liver glucose protects them; on thawing the heart returns first. Chickadees cache food and use the hippocampus to find it; one study found more new neurons there in fall, but later studies disagree and the cells seem to replace old ones, so we never say they grow bigger brains. In most small mammal hibernators, hibernation is long torpor bouts with brief rewarming, often fuelled by brown fat; a bear cools only a few degrees. With no brood, honeybee clusters keep the core above about 64 F. Day length is thought to trigger the hare molt (about 40 days). About 6 in of snow keeps the subnivean zone near 32 F. Woolly bears make glycerol; we skip the stripe folklore. Watch wildlife from a distance; use warm, not hot, water.